Supplementary Material S6
Claim, Citation and Evidence Matrix

Facial Emotion Recognition through Machine Learning in Children with Autism Spectrum Disorder: A systematic scoping review of open-access computational evidence and of the gaps toward therapeutic monitoring

This document responds to observation I-08 of the advisory review of 4 November 2026, which asks for every substantive claim of the Discussion and the Conclusions to be traced to its citation and to the evidence that supports it. Each row gives the section of the manuscript, the claim as stated before this audit, the citations or tables it rests on, the evidence located in the repository or in the manuscript, and the status.

The audit covers every sentence of Sections 6 and 7 that states a number, cites a study or uses evaluative, causal or transferability wording, and the one sentence of Section 1 with confirmatory wording. Evidence is located in the extraction matrix (`data/extraction_master.csv`, column named in each row), in the tables of the manuscript and in Supplementary Materials S3 and S4.

Status: *Supported*, the evidence matches the claim; *Corrected*, the wording or the citations were changed in the manuscript so that the claim matches the evidence, and the evidence column states the change; *Fit check pending*, the claim is consistent with the extracted data but the fit of a citation or a page-level source still has to be checked against the full text.

Totals: 52 claims; 32 supported, 15 corrected and 5 with a fit check pending. In all, 10 rows, 3 of them corrected, still need a citation or page-level check against the full text.

#	Sec.	Claim	Citations or table	Evidence and location	Status
1	6.1	CNN / Deep Learning is the family of 69.6% of the 23 studies with an identifiable ML architecture.	Table 10	`model_family`: CNN / Deep Learning in 16 of 23.	Supported
2	6.1	The predominance of CNN reflects the convergence of practical accessibility and methodological maturity.	None	No extracted variable supports an explanation. Reworded as a hypothesis that the corpus cannot test.	Corrected
3	6.1	The highest accuracies (>95%) come from hybrid or ensemble CNN configurations on images without clinical traceability.	Aljbori2025, Hosney2024, Vedantham2025, Alam2025	Table 12: 99.01%, 97.2%, 96.5%, 96.0%. `dataset_category`: Kaggle ASD collections. S4 C.2: 3D-CNN + LSTM, YOLO + CNN, CNN hybrid with GWO, CNN ensemble.	Supported
4	6.1	Hassan et al. report 95.6% (ASD level, DREAM) and Hwang et al. 88.0% (developmental disability screening, own clinical data), both in D4.	Hassan2025, Hwang2024	Table 12; `domain` = D4; `dataset_category`: DREAM and Own clinical data.	Supported
5	6.1	17 of the 31 eligible studies are from 2024 and 2025; none is from 2023.	Table 21	`year`: 2024 (4), 2025 (13), 2023 (0).	Supported
6	6.1	The studies with the highest documented ecological validity integrate temporal modelling or multimodal fusion in real therapeutic contexts.	Hwang2024, Hassan2025, Kiejdo2025	`lim_ecological` = 0 in the three. S4 C.2: LSTM + 2D-FAN, RF + XGBoost on DREAM (three or more sources, Table 16), late fusion of BiLSTM in robot sessions. Hwang et al. is a screening context, not a therapeutic one: reworded to clinical or therapeutic contexts.	Corrected
7	6.2	The six studies on web repositories without ethical oversight report 99.01%, 97.2%, 96.5%, 96.0%, 88.3% and 84.0%.	Aljbori2025, Hosney2024, Vedantham2025, Alam2025, Kurian2024, Shaik2025	Table 12; `dataset_category`: Kaggle ASD collections (5) and Web videos (1).	Supported
8	6.2	Clinical-origin studies report lower values within each domain: 79.1% (66.9% on full CAFE), 90.7% with 33 children, AUROC 0.86 to 0.92 with 4 children, 88.0% with 89 children and 88.7% with 81 children.	Washington2022, Wang2025, Shi2022, Hwang2024, Chen2024	Table 12 and its notes (Sub. A, f, i); S4 Section A, note 3.	Supported

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#	Sec.	Claim	Citations or table	Evidence and location	Status
9	6.2	The 67.8% of Johnson et al. is measured on AffectNet, not on the participants with ASD.	Johnson2025	Table 12, note a; S4 note i.	Supported
10	6.2	Of the five highest accuracies, four come from web repositories and the fifth (Fan et al.) measures specialist-reviewed feedback.	Fan2025, Shaik2025, Hwang2024, Chen2024	Table 12 ranking: 99.01%, 97.6%, 97.2%, 96.5%, 96.0%; S4 note p.	Supported
11	6.2	Hassan et al. reach 95.6% on DREAM, a public dataset curated with ADOS diagnosis and documented parental consent.	Hassan2025	Table 12, note c; S3 Section 7; S4 note l.	Supported
12	6.2	Han et al. reach 95.6% with 90 children by fusing EEG and eye-tracking in a laboratory.	Han2022	Table 12; S4 note q.	Supported
13	6.2	Aljbori et al. report 99.01% in the results table and 99.82% in the abstract; Hosney et al. use a dataset created by a co-author; El Rhatassi et al. fall from 94.5% to 53.0%.	Aljbori2025, Hosney2024, Rhatassi2025	S4 notes g and d; Table 12 note; S4 Section B.	Supported
14	6.2	ACC is reported numerically in 20 of 22 studies with supervised metrics (91%), F1 in 11, Recall in 9, AUC in 7 and Precision in 6.	Table 11	`supervised_metrics` = 1 in 22; `acc` 20, `f1` 11, `recall` 9, `auc` 7, `precision` 6.	Supported
15	6.2	This distribution can be explained by researchers' familiarity with standard metrics.	None	No study states this reason. Reworded as a hypothesis that the review does not test.	Corrected
16	6.2	AUC improves on accuracy but is insufficient alone under severe class imbalance.	Radocaj2025, Hosney2024	Methodological argument of Section 4.2; both studies recognise class imbalance (Table 8, row 5).	Supported
17	6.2	The most used public dataset is the Kaggle autism collections; the earlier text also named FER-2013.	Table 13	`dataset_category`: Kaggle ASD collections 5 of 31; FER-2013, FERAC, CAFE, DREAM and web videos 1 each. FER-2013 is not among the most used: corrected.	Corrected
18	6.2	Private datasets account for 67.7% of the eligible corpus.	Table 13	`dataset_category`: own clinical 13, clinical eye-tracking 5, robot therapy 2, own non-ASD 1; 21 of 31.	Supported
19	6.2	No public dataset with Latin American paediatric ASD population was identified in the retrieved evidence.	None	S3 Sections 2 to 12: no dataset documents a Latin American population. The claim is limited to the retrieved evidence.	Supported
20	6.2	Declared limitations: imbalance and sampling 27 of 31, recording and annotation 11 of 31, ecological 10 of 31.	Table 17	`lim_imbalance_sampling` 27, `lim_annotation` 11, `lim_ecological` 10.	Supported
21	6.2	Recording inconsistency is due to annotation errors, non-standard protocols and inter-rater variability.	None	Causal wording without extracted evidence. Reworded as the definition of the category declared by the studies.	Corrected
22	6.2	Negative or not determinable MMAT judgements concentrate in representativeness (17 of 31) and confounding (18 of 31).	Figure 2	`mmat_representativeness`: N 9, ND 8; `mmat_confounding`: N 7, ND 11.	Supported
23	6.2	Four studies are negative in both domains: three on web repositories and Fan et al.	Aljbori2025, Hosney2024, Vedantham2025, Fan2025	N in both MMAT columns for exactly these four.	Supported
24	6.2	Seven studies meet the five domains of their category.	He2019, Washington2022, Radocaj2025, Hwang2024, Hassan2025, Leo2018, Wang2025	Y in the five MMAT columns for exactly these seven.	Supported
25	6.2	Values from web-repository studies have no evidence supporting transferability to clinical practice.	Aljbori2025, Hosney2024	MMAT N in representativeness and confounding; Table 5 (no consent, no verified diagnosis).	Supported
26	6.3	Representation: facial or gaze descriptors 16 of 31, end-to-end CNN 12, physiological signals 2, capture format 1.	Tables 14 and 15	`input_group`: 16, 12, 2, 1.	Supported
27	6.3	Two further studies combine face or gaze with electrodermal activity or EEG.	Kiejdo2025, Han2022	Table 16 note (bimodal list); S4 C.2.	Supported

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#	Sec.	Claim	Citations or table	Evidence and location	Status
28	6.3	No study reports session-to-session tracking; Hassan et al. (61 children, 3,121 sessions), Shi et al. (4 children, 19 sessions) and four pre-post designs do not meet the definition.	Hassan2025, Shi2022, Lecciso2021, Voss2019, Fan2025, Wall2025	S4 C.1 and notes e, f and r. Page-level source per study pending (I-11).	Supported
29	6.3	No study declares which error it prioritises or reports performance disaggregated by skin tone, sex, severity or comorbidity.	Table 5	Absence recorded during extraction; no study coded with these variables. Page-level verification pending with the full texts.	Fit check pending
30	6.3	Table 22 synthesises the 18 eligible studies with verifiable data provenance.	Table 22	`translational` = 1 in 18.	Supported
31	6.3	The absence of tools designed for monitoring during interventions is the most significant gap for the Latin American context.	Shaik2025, Hassan2025	S4 C.2: binary ASD/TD and ASD level, examples of the diagnostic orientation.	Supported
32	7.1	None of the 31 studies reports the variables that operationally define therapeutic monitoring.	Section 1	S4 C.1: intraindividual series, progress outcomes, test-retest and responsiveness are No or ND in all 31.	Supported
33	7.1	Primary evidence: 14 studies (D1 10, D2 4); D3 and D4 (17) are comparative context.	Section 2.8	`domain`: D1 10, D2 4, D3 6, D4 11; `primary_evidence` = 1 in 14.	Supported
34	7.1	Hybrid CNN + LSTM and fusion architectures appear mainly in clinical, robot-therapy and mobile studies; single CNN with transfer learning predominates on public benchmarks.	None	Hybrid part: S4 C.2 lists Hwang, Chen, Kiejdo (clinical, mobile or robot) and Aljbori (web), 3 of 4. Transfer-learning part: the per-study assignment to Table 14 is not yet released (I-03).	Fit check pending
35	7.1	The highest accuracies of the corpus (>95%) come from web-repository studies.	Fan2025	Han et al. (95.6%), Hassan et al. (95.6%) and Fan et al. (97.6%) also exceed 95%. Corrected to four of the five highest, with Fan et al. named.	Corrected
36	7.1	Clinical-origin reference ranges: D1 68.0% to 90.7%, D2 AUROC 0.86 to 0.92, D4 88.0% to 95.6%.	Pliska2025	Table 12: Pliska et al. (D4, clinical) reports 81.3% without validation on unseen data. Corrected: the D4 range is stated for studies with validation and the 81.3% is named.	Corrected
37	7.1	These models face critical interpretability and phenotypic-heterogeneity challenges.	Hassan2025, Shi2022 replaced by Johnson2025, Alam2025, Shi2022, Chen2024	Table 8: interpretability recognised by Johnson and Alam; generalisation in small samples by Shi and Chen, among others. Hassan et al. recognise neither. Citations corrected.	Corrected
38	7.1	Only two studies report balanced accuracy and only two report confidence intervals.	Washington2022, Kiejdo2025, Wang2025, Carpenter2020	Table 12 notes e, f and m; Sub. A.	Supported
39	7.1	No evidence allows metric standards for session-to-session monitoring to be set.	Radocaj2025	Absence claim supported by S4 C.1. The fit of the citation is to be checked against the full text.	Fit check pending
40	7.1	DREAM: 61 children aged 3 to 6 with ADOS diagnosis, 3,121 sessions, 306 hours, CC BY-NC-SA 4.0, without facial images.	Hassan2025	S3 Section 7.	Supported
41	7.1	74.2% of the corpus is unimodal and 3 of 31 integrate three or more sources.	Table 16	Table 16; the three and the five bimodal studies are named in its note. Per-study coding of the other dimensions pending (I-03).	Supported
42	7.1	Fusion of video, physiological and contextual data could improve robustness.	Leo2018, Hassan2025; Han2022 added	Han et al. report 86.67% (eye-tracking) and 81.11% (EEG) against 95.56% fused (S4 note q). Han2022 added as evidence; the fit of Leo2018 is to be checked against the full text.	Corrected
43	7.1	Balancing and partition practice: 2 of 23 declare a balancing technique and 17 omit it; fixed split 12, K-fold 6, no partition 2; 10 of 23 declare no hyperparameter strategy.	Tables 19 and 20	Per-study citations in Tables 19 and 20.	Supported
44	7.1	Appropriate balancing can generate substantial improvements in Recall and F1 without changing the architecture.	Chawla2002, He2008 added	No corpus study quantifies it. Reworded as an improvement on the minority class and supported with the methodological sources of SMOTE and ADASYN.	Corrected

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45	7.2	Early intervention is the stage in which intensive support has the greatest long-term impact.	None	Uncited clinical claim outside the scope of the review. Reworded as a recognised priority, stating that the review does not assess that evidence.	Corrected
46	7.2	Method repertoire: LIME, SHAP, Grad-CAM, MRMR, SMOTE and ADASYN.	Ribeiro2016, Lundberg2017, Selvaraju2017, Peng2005, Chawla2002, He2008	Primary methodological sources of each method.	Supported
47	7.2	Transcultural validation in Latin American paediatric populations is required.	Shaik2025, Washington2022	Prospective statement. The fit of both citations is to be checked against the full texts.	Fit check pending
48	7.2	Longitudinal ecological momentary assessment could provide unprecedented information.	Johnson2025, Shi2022	"Unprecedented" removed. Prospective statement; citation fit to be checked against the full texts.	Corrected
49	7.2	XAI systems should support, not replace, clinical judgement.	Radocaj2025	Prospective statement. The fit of the citation is to be checked against the full text.	Fit check pending
50	7.2	Reinforcement learning could enable truly personalised intervention trajectories.	None	"Truly" removed; prospective statement without a corpus claim.	Corrected
51	7.2	Multimodal Transformers can capture subtle patterns undetectable by unimodal analysis.	Washington2022	Reworded as a hypothesis not tested with therapeutic-session data. Citation fit to be checked against the full text.	Corrected
52	1	The review by Baños et al. confirms that the face is the predominant modality.	Banos2024	"Confirms" replaced by "reports": the statement summarises another review and is not verified by this one.	Corrected

Notes.

Section numbers follow the manuscript: 1, Introduction; 6.1 to 6.3, Discussion; 7.1 and 7.2, Conclusions and future research.

Citation keys are those of the bibliography of the manuscript. Every citation added or replaced in this audit is named in the citations column of its row.

Rows with a fit check pending require the full text of the cited study; they are resolved together with observations I-01 and I-11.